

ALGORITHMS FUNDAMENTALS

Introduction, Design & Types

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Course Outline

1 Introduction to Algorithms

2 Algorithm Design

3 Algorithm Types



Outline

1 Introduction to Algorithms

2 Algorithm Design

3 Algorithm Types



What is an Algorithm?

Definition

An algorithm is a finite sequence of well-defined instructions that can be mechanically executed to solve a computational problem or perform a task.

Key Components:

- **Input:** Zero or more quantities supplied externally
- **Output:** One or more quantities produced
- **Instructions:** Precise and unambiguous steps
- **Execution:** Can be performed mechanically

repetition

automation

no ambiguity

Remember

An algorithm is a *method* for solving problems, not the implementation itself!



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theoretical

algorithm $\neq$ code



Algorithm Characteristics

Essential Properties

- 1 **Finiteness:** Must terminate after finite number of steps
- 2 **Definiteness:** Each step must be precisely defined
- 3 **Input:** Zero or more input values
- 4 **Output:** One or more output values
- 5 **Effectiveness:** Operations must be basic and feasible

Example:

Making Coffee Algorithm:

- 1 Boil water (definite action)
- 2 Add coffee grounds (precise amount)
- 3 Wait 4 minutes (finite time)
- 4 Pour into cup (effective operation)

while 1:

1

2

3

4

Algorithm Characteristics

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- ① **Finiteness**: Must terminate after finite number of steps
- ② **Definiteness**: Each step must be precisely defined
- ③ **Input**: Zero or more input values — *Coffee, water, mug*
- ④ **Output**: One or more output values — *cup*
- ⑤ **Effectiveness**: Operations must be basic and feasible

Example

Making Coffee Algorithm:

- ① Boil water (definite action)
- ② Add coffee grounds (precise amount)
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Algorithm vs. Program vs. Process

Algorithm

- Abstract concept
- Language-independent
- Mathematical description
- Design phase

Program

- Concrete implementation
- Language-specific
- Code representation
- Implementation phase

Process

- Runtime execution
- System-dependent
- Active computation
- Execution phase

Example

Finding the maximum number.

- **Algorithm:** "Compare each element with current maximum"
- **Program:** Python code implementing the comparison
- **Process:** Program running in memory, using CPU cycles

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Types of Algorithmic Thinking

Fundamental Approaches

- ➊ **Sequential Thinking:** Step-by-step execution
- ➋ **Conditional Thinking:** Decision-based branching
- ➌ **Iterative Thinking:** Repetitive operations
- ➍ **Recursive Thinking:** Self-referential solutions
- ➎ **Parallel Thinking:** Concurrent execution

Example (Daily Life)

- Getting dressed (sequential)
- Choosing route to university (conditional)
- Studying until understanding (iterative)
- Recalling a recipe (recursive)
- Cooking multiple dishes (parallel)

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Study Case: Euclidean GCD Algorithm

Greatest Common Divisor Problem

Input: Two positive integers a and b

Output: The largest integer that divides both a and b

Algorithm 1 Euclidean GCD Algorithm

Input: Integers a, b where $a \geq b > 0$

while $b \neq 0$ **do**

$r \leftarrow a \bmod b$

$a \leftarrow b$

$b \leftarrow r$

end while

return a



Study Case: Euclidean GCD Algorithm

Greatest Common Divisor Problem

Input: Two positive integers a and b

Output: The largest integer that divides both a and b

Algorithm 2 Euclidean GCD Algorithm

Input: Integers a, b where $a \geq b > 0$

while $b \neq 0$ **do**

$r \leftarrow a \bmod b$

$a \leftarrow b$

$b \leftarrow r$

end while

return a

*In Py for
notebooks*

Define f as a function



Exercise Time! [1]

Exercise 1: Square Root Calculation

Design an algorithm to calculate the square root of a positive number using the Babylonian method:

- 1 Start with an initial guess x_0 (e.g., $S/2$ for number S)
- 2 Improve the guess: $x_{n+1} = \frac{1}{2}(x_n + \frac{S}{x_n})$
- 3 Repeat until convergence

Exercise 2: Find Maximum in an Array

Write an algorithm that finds the maximum element in an array of integers.

Consider: What if the array is empty? What about duplicate maximums?



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$E \rightarrow in$

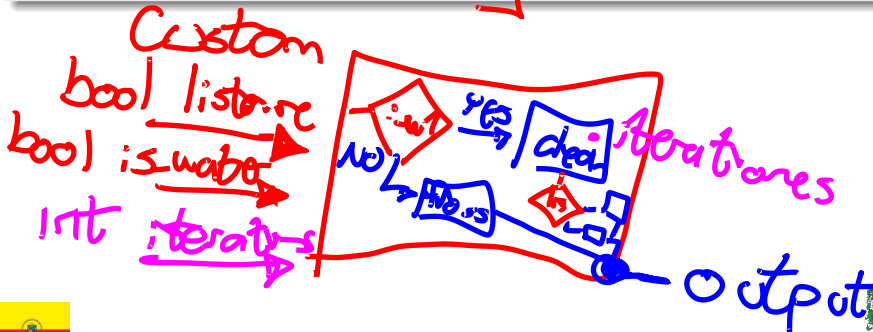


Exercise Time! [2]

Exercise 3: Algorithms in Daily Life

Identify and describe three algorithms you use in your daily life. For each:

- Define clear inputs and outputs
- List the precise steps
- Verify all algorithm characteristics



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Mathematical Notation for Algorithms [I]

Mathematical Notation

Mathematical notation provides a formal way to express algorithms using symbols, formulas, and structures from mathematics.

Why Mathematical Notation?

- **Precision:** Unambiguous specification
- **Universality:** Language-independent
- **Conciseness:** Compact representation
- **Analysis:** Enables formal verification

↪ Universal

factorial(n) = $n * \text{factorial}(n-1)$
 if $n > 1$,
 otherwise

```
def factorial(n):
    if n > 1:
        return n * factorial(n-1)
    return 1
```



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$$f(4) \rightarrow 4 \cdot f(3)$$

$$\quad \quad \quad 3 \cdot f(2)$$

$$\quad \quad \quad 2 \cdot f(1)$$

$$\text{output} = 24$$



Mathematical Notation for Algorithms [II]

Set Operations

- $S = \{x \in \mathbb{N} : x \text{ is prime and } x < 20\}$
- $\forall x \in S, P(x)$ (for all elements in S, property P holds)
- $\exists x \in S : Q(x)$ (there exists an element in S with property Q)

def is_prime(n):

$S = []$ $f(a, b) = \begin{cases} a & \text{if } b = 0 \\ f(b, a \bmod b) & \text{otherwise} \end{cases}$

for x in range(1, 20):
 if is_prime(x):
 S.add(x)



Mathematical Notation for Algorithms [II]

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Example

Function Definition:

recursive

$$f : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$$

f?

$$f(a, b) = \begin{cases} a & \text{if } b = 0 \\ f(b, a \bmod b) & \text{otherwise} \end{cases}$$

base case

recursive case

Euclidean GCD



Converting Mathematical Formulas to Pseudocode

Mathematical Expression

$$\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$$

Another Example

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

```

1: sum ← 0
2: for i = 1 to n do
3:   sum ← sum + i2
4: end for
5: formula ←  $\frac{n(n+1)(2n+1)}{6}$ 
6: if sum = formula then
7:   return true
8: else
9:   return false
10: end if

```

```

1: result ← 1
2: term ← 1
3: n ← 1
4: while |term| >  $\epsilon$  do
5:   term ←  $\frac{\text{term} \cdot x}{n}$ 
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9: return result

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Another Example

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

Handwritten annotations:

- fact* ← 1 (blue)
- Convergence* (pink arrow pointing to the while loop)
- fact* (blue, pointing to the term calculation)
- fact = fact * n* (blue, pointing to the update of *fact*)
- fact* (blue, pointing to the final return)

```

1: result ← 1
2: term ← 1
3: n ← 1
4: while |term| >  $\epsilon$  do
5:   term ←  $\frac{x^n}{\text{fact}}$ 
6:   result ← result + term
7:   n ← n + 1
8: end while
9: return result

```

Study Case: Factorial Using Different Notations

Mathematical

recursive

$$n! = \begin{cases} 1 & \text{if } n = 0 \\ n \cdot (n-1)! & \text{if } n > 0 \end{cases}$$

<=

$$n! = \prod_{i=1}^n i$$

Iterative Pseudocode

```

1: function FACTORIAL(n)
2:   result  $\leftarrow$  1
3:   for i = 1 to n do
4:     result  $\leftarrow$  result  $\times$  i
5:   end for
6:   return result
7: end function

```

def factorial(n):

if n = 0:
return 1

else:
*return n * factorial(n-1)*



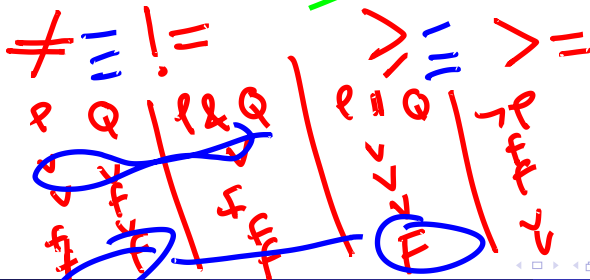
Pseudocode Standards and Conventions

Basic Elements

- **Keywords:** if, then, else, while, for, return
- **Assignment:** $x \leftarrow \text{value}$ or $x = \text{value}$
- **Comparison:** =, ≠, <, >, ≤, >
- **Logic:** and, or, not
- **Comments:** // Single line, /* Multi-line */

palabras reservadas

memory



Algorithm Documentation Best Practices

Essential Documentation Elements

- ① **Purpose:** What **does** the algorithm do?
- ② **Preconditions:** What must be true **before** execution?
- ③ **Postconditions:** What is guaranteed **after** execution?
- ④ **Parameters:** Input and **output** specifications
- ⑤ **Complexity:** Time and **space** requirements
- ⑥ **Examples:** **Sample** inputs and outputs



Study Case: Fibonacci Sequence

Problem Definition

The Fibonacci sequence: $F_0 = 0, F_1 = 1, F_n = F_{n-1} + F_{n-2}$ for $n \geq 2$

Purpose: Compute the n th Fibonacci number

Preconditions: $n \geq 0$ (non-negative integer)

Postconditions: Returns F_n where $F_0 = 0, F_1 = 1$

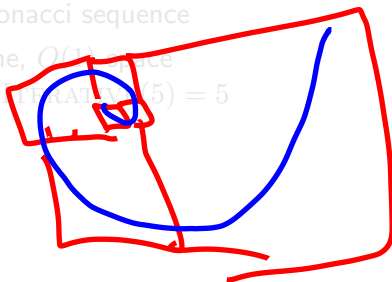
Parameters:

- n : position in Fibonacci sequence

Complexity: $O(n)$ time, $O(1)$ space

Example: FIBONACCI(5) = 5

0 1 1 2 3 5 8 13 21



Study Case: Fibonacci Sequence

Problem Definition

The Fibonacci sequence: $F_0 = 0, F_1 = 1, F_n = F_{n-1} + F_{n-2}$ for $n \geq 2$

Purpose: Compute the nth Fibonacci number

Preconditions: $n > 0$ (non-negative integer)

Postconditions: Returns F_n where $F_0 = 0, F_1 = 1$

Parameters:

- n : position in Fibonacci sequence

Complexity: $O(n)$ time, $O(1)$ space

Example: FIBONACCIITERATIVE(5) = 5



Control Structures

Sequential

- 1: Step 1
- 2: Step 2
- 3: Step 3
- 4: ...

Execution: One after another

Conditional

- 1: **if** condition **then**
- 2: Action A
- 3: **else**
- 4: Action B
- 5: **end if**

Execution: Based on condition

Iterative

- 1: **while** condition **do**
- 2: Action
- 3: **end while**
- 4: **for** $i = 1$ to n **do**
- 5: Action
- 6: **end for**

Execution: Repeated



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Control Flow Concepts

- **Entry Point:** Where **execution begins**
- **Exit Point:** Where **execution ends**
- **Decision Points:** Where **flow branches**
- **Loop Control:** How **iterations** are managed



Study Case: Decision-Making Algorithms

Grade Classification Algorithm

Problem: Convert numerical grade to letter grade

Decision Tree

- $Score \geq 90? \rightarrow A$
- $Score \geq 80? \rightarrow B$
- $Score \geq 70? \rightarrow C$
- $Score \geq 60? \rightarrow D$
- $Otherwise \rightarrow F$

Key Concepts

- **Mutually exclusive** conditions
- **Order matters** in if-elsif chains
- **Default case** handling



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Greedy Algorithms: Philosophy and Approach

Definition

A **greedy algorithm** makes locally optimal choices at each step, hoping to find a global optimum.

Greedy Strategy

- 1 **Greedy Choice:** At each step, choose the **best available** option
- 2 **Optimal Substructure:** Optimal solution contains optimal solutions to subproblems
- 3 **No Backtracking:** Never reconsider previous choices



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Greedy Algorithms: *When to Use Them*

When Greedy **Works**:

- Activity selection
- Minimum spanning trees
- Huffman coding

When Greedy **Fails**:

- 0/1 Knapsack problem
- Traveling salesman
- Graph coloring



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Study Case: Coin Change Problem

Problem Statement

Given: Coin denominations $[d_1, d_2, \dots, d_k]$ and amount n

Goal: Find minimum number of coins to make amount n



Study Case: Coin Change Problem [Solution]

Denominations: [25, 10, 5, 1] cents, **Amount:** 67 cents

```
1: Algorithm GREEDYCOINCHANGE(denominations, amount)
2: result  $\leftarrow []$ 
3: for each denomination d in descending order do
4:   while amount  $\geq d$  do
5:     result.append(d)
6:     amount  $\leftarrow$  amount  $- d$ 
7:   end while
8: end for
9: return result
```

Solution: $67 = 25 + 25 + 10 + 5 + 1 + 1$ (6 coins)



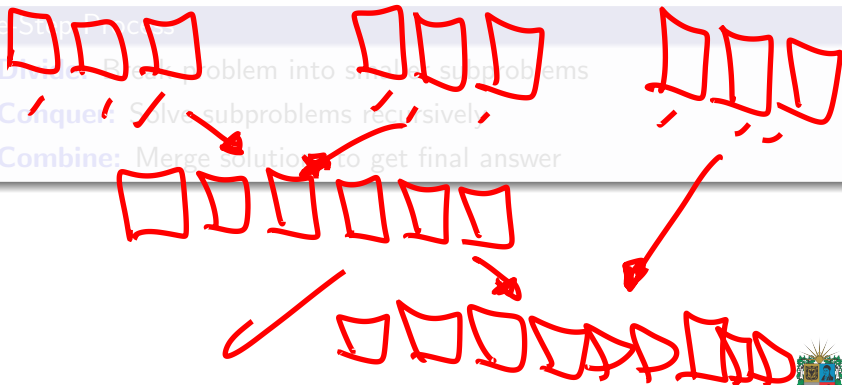
Divide and Conquer: Methodology

Definition

Divide and Conquer breaks a problem into smaller subproblems, solves them recursively, then combines solutions.

Three Steps Process

- ① **Divide:** Break problem into smaller subproblems
- ② **Conquer:** Solve subproblems recursively
- ③ **Combine:** Merge solutions to get final answer



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Definition

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Three-Step Process

- 1 **Divide:** Break problem into smaller subproblems
- 2 **Conquer:** Solve subproblems **recursively**
- 3 **Combine:** Merge solutions to get final answer



Divide and Conquer: Algorithm Template

```
1: Algorithm DIVIDEANDCONQUER(problem)
2: if problem is small enough then
3:   return SOLVEDIRECTLY(problem)
4: else
5:   subproblems  $\leftarrow$  DIVIDE(problem)
6:   for each subproblem do
7:     solution  $\leftarrow$  DIVIDEANDCONQUER(subproblem)
8:   end for
9:   return COMBINE(solutions)
10: end if
```

Handwritten annotations: A pink oval highlights the function name DIVIDEANDCONQUER and its argument *problem*. A green bracket on the right side of lines 2-3 is labeled **CONQUER** in green. A red arrow points from the **return** statement in line 7 to the **CONQUER** label. A blue oval highlights the **return** statement in line 9 and the argument *solutions*.



Problem Decomposition Strategies

- **Top-Down:** Start with the main problem and break it down into subproblems
- **Bottom-Up:** Solve smaller subproblems first and use their solutions to build up to the main problem
- **Recursive:** Define the problem in terms of smaller instances of itself



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□ = solve



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Handwritten recursive formula: $f(x) = \begin{cases} v, & \text{if } x \text{ is } 1 \\ f(x-1) + v, & \text{if } x > 1 \end{cases}$

Annotations: A red circle highlights $f(x)$. A green circle highlights x . A red circle highlights $f(x-1)$. A green circle highlights $x-1$. A red arrow points from the red circle to the word "smaller" written in green. A green arrow points from the green circle to the word "smaller". A magenta arrow points from the word "Recursive" in the list above to the formula.



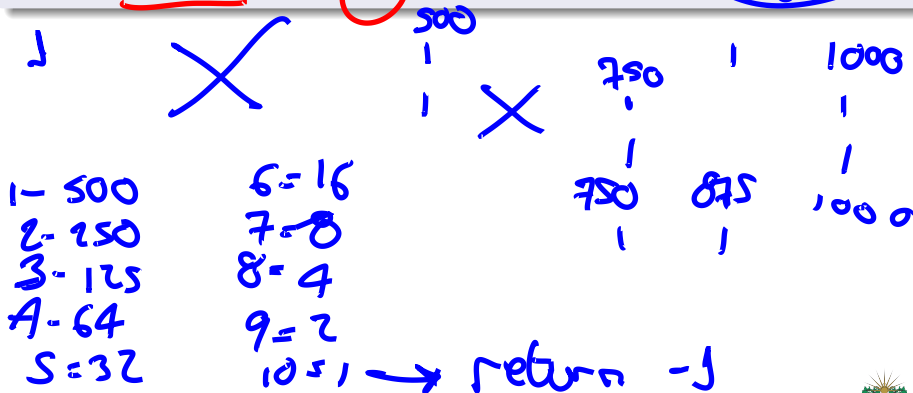
Study Case: Binary Search

Problem

Input: Sorted array $A[1..n]$ and search value x

Output: Index of x in A , or -1 if not found

896



Study Case: Merge Sort

Problem

Input: Array $A[1..n]$ of comparable elements

→ numbers

Output: Array sorted in ascending order



Demo Time! Clean the House Exercise •

Problem

You need to clean your house efficiently. You have different rooms and different cleaning tasks.

Greedy Strategy: Always clean the dirtiest room first

- 1 Assess dirt level of all rooms
- 2 Choose room with highest dirt level
- 3 Clean that room completely
- 4 Repeat until all rooms clean

Advantage: Maximum immediate impact

Divide & Conquer Strategy:
Split house systematically

- 1 Divide house into sections (floors/wings)
- 2 Recursively clean each section
- 3 Within each room, divide into areas
- 4 Clean areas systematically

Advantage: Systematic coverage



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Exercise: Knapsack Problem

Problem Statement

Given: Knapsack capacity W items with weights w_i and values v_i

Goal: Maximize total value without exceeding weight capacity

Greedy $\Rightarrow$ sort by value

$v_k, w_k, \dots, v_n, w_n$
 $\downarrow$ if $w_k < \text{available } W$



Exercise: Knapsack Problem [Solution]

0/1 Knapsack (Greedy Approach)

Cannot take fractions - either take entire item or leave it

- **Strategy:** Sort by value/weight ratio, take whole items
- **Optimal:** No! (Greedy doesn't guarantee optimal solution)

- 1: Sort items by v_i/w_i descending
- 2: **for** each item i **do**
- 3: **if** $w_i \leq$ remaining capacity **then**
- 4: Take entire item
- 5: Update remaining capacity
- 6: **end if**
- 7: **end for**



Brute Force: Exhaustive Search

Definition

Brute force algorithms try all possible solutions until finding the correct one.

Characteristics:

- **Exhaustive:** Examines every possibility
- **Guaranteed:** Always finds optimal solution (if exists)
- **Expensive:** Often exponential time complexity
- **Simple:** Easy to understand and implement



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- **Simple:** Easy to **understand and implement**

$$45 \times 44 \times 43 \times 42 \times 41 \times 40 = \frac{45!}{39!}$$

$$\approx 5.8 \times 10^9$$



Brute Force: When to Use It

- ① Problem size is small
- ② No efficient algorithm is known
- ③ Correctness is more important than efficiency
- ④ As baseline for comparing other algorithms
- ⑤ When optimization overhead exceeds brute force cost

ChatGPT

Knapsack

BF	Greedy	D&Q
200	190	180
	100	

4.5
20000 30000
GPUS
~3 weeks
200M USD
↑ trillion



Study Case: Password Cracking

Problem

Given: Encrypted password hash and character set

Goal: Find the original password

$$\begin{array}{l}
 \underline{\underline{a-z}} \dots \dots \dots a-z \dots \\
 28^8 = 4.8 \times 10^8 \\
 \underline{\underline{0-9a-zA-Z}} \dots \dots \dots = 3.7 \times 10^{11} \quad / \quad \text{all possible}
 \end{array}$$



Study Case: Password Cracking [Solution]

```

1: Algorithm BRUTEFORCEPASSWORD(hash, maxlength)
2: for length = 1 to maxlength do
3:   for each possible string s of given length do
4:     if HASH(s) = hash then
5:       return s // Password found!
6:     end if
7:   end for
8: end for
9: return null // Password not found
  
```

Handwritten notes: "brute force" written vertically in pink on the right side of the algorithm.

Character set: [a-z] (26 characters)

Password length: 8 characters

Total attempts: $26^8 \approx 208$ billion

Time estimate: Weeks on single computer!



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base exponent



Backtracking: Systematic Trial and Error

Definition

Backtracking explores solution space systematically, **abandoning partial solutions** that cannot lead to valid solutions.

Backtracking Strategy

- 1 **Choose:** Make a choice from available options
- 2 **Explore:** Recursively explore consequences
- 3 **Unchoose:** If path fails, backtrack and try another



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Backtracking: Algorithm

```

1: Algorithm BACKTRACK(partial_solution)
2: if ISCOMPLETE(partial_solution) then
3:   return partial_solution
4: end if
5: for each possible choice c do
6:   if ISVALID(partial_solution, c) then
7:     partial_solution.add(c)
8:     result  $\leftarrow$  BACKTRACK(partial_solution)
9:     if result  $\neq$  null then
10:      return result
11:    end if
12:    partial_solution.remove(c) // Backtrack!
13:  end if
14: end for
15: return null
  
```



Study Case: N-Queens Problem

Problem

Place N queens on an $N \times N$ chessboard such that no two queens attack each other.



Study Case: N-Queens Problem [Solution]

```

1: Algorithm SOLVENQUEENS(board, row)
2: if row = N then
3:   return board // Solution found!
4: end if
5: for each column col in 0 to N - 1 do
6:   if ISSAFE(board, row, col) then
7:     board[row][col] ← Q // Place queen
8:     result ← SOLVENQUEENS(board, row + 1)
9:     if result ≠ null then
10:      return result
11:    end if
12:    board[row][col] ← . // Remove queen (backtrack)
13:  end if
14: end for
15: return null // No solution found

```



Implementation Patterns for Backtracking

- **Recursive Structure:** Backtracking algorithms are typically implemented using recursion.
- **State Representation:** Maintain a representation of the current state (e.g., partial solution).
- **Choice Points:** Identify points where decisions are made (choices to explore).
- **Validity Checks:** Implement checks to determine if the current state is valid.
- **Base Case:** Define a base case for when a complete solution is found.
- **Backtracking Mechanism:** Ensure that the algorithm can revert to previous states when necessary.



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Summary: Algorithm Types

Key Takeaways

- **Greedy:** Make locally optimal choices (works when greedy choice property holds)
- **Divide & Conquer:** Break into subproblems, solve recursively, combine solutions
- **Brute Force:** Try all possibilities (guarantees correctness but expensive)
- **Backtracking:** Systematic search with pruning (intelligent brute force)



Summary: Algorithm Types

Algorithm Selection Guide

- **Small problem size?** → Consider brute force
- **Optimization problem with greedy choice property?** → Try greedy
- **Problem naturally divides into subproblems?** → Use divide & conquer
- **Constraint satisfaction or search problem?** → Apply backtracking



Outline

- 1 Introduction to Algorithms
- 2 Algorithm Design
- 3 Algorithm Types



Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/computer-sciences-i>

